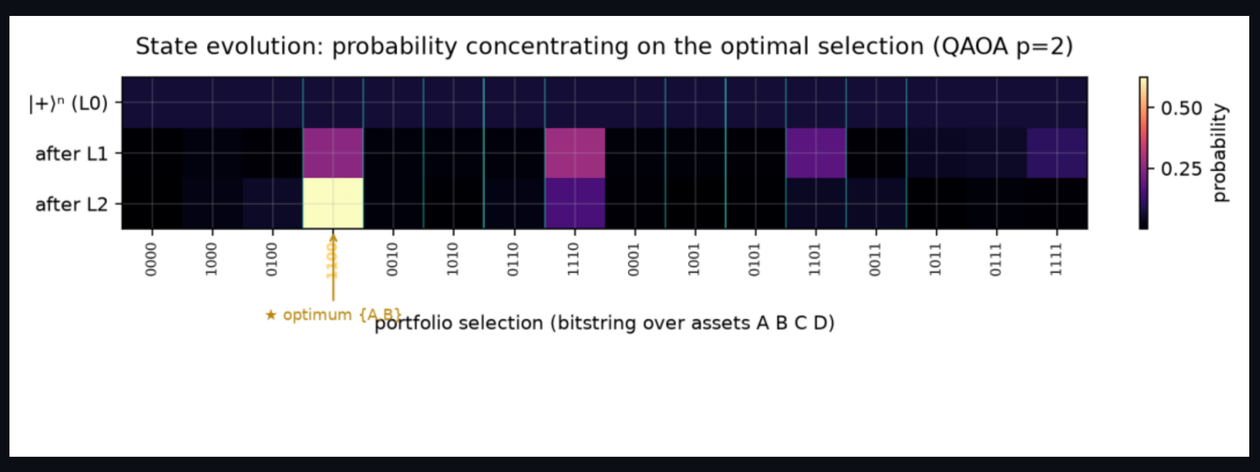
Watching Language Steer a Quantum Optimizer

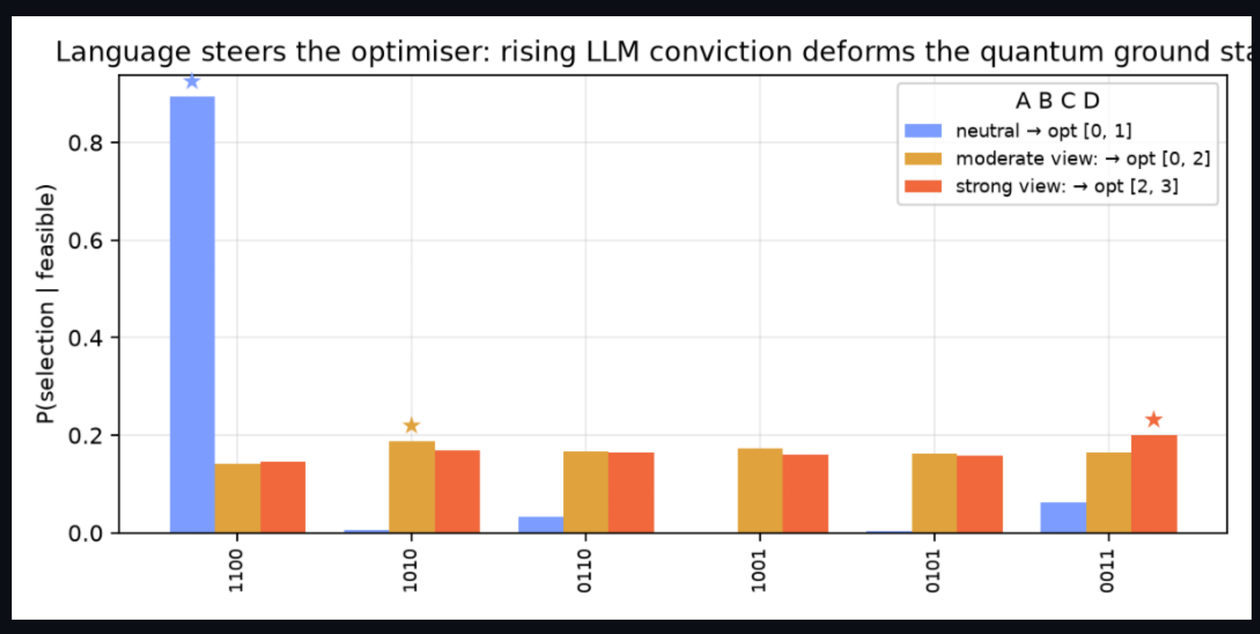
State-Evolution Visualization of LLM-Conditioned QAOA for Cardinality-Constrained Portfolio Selection

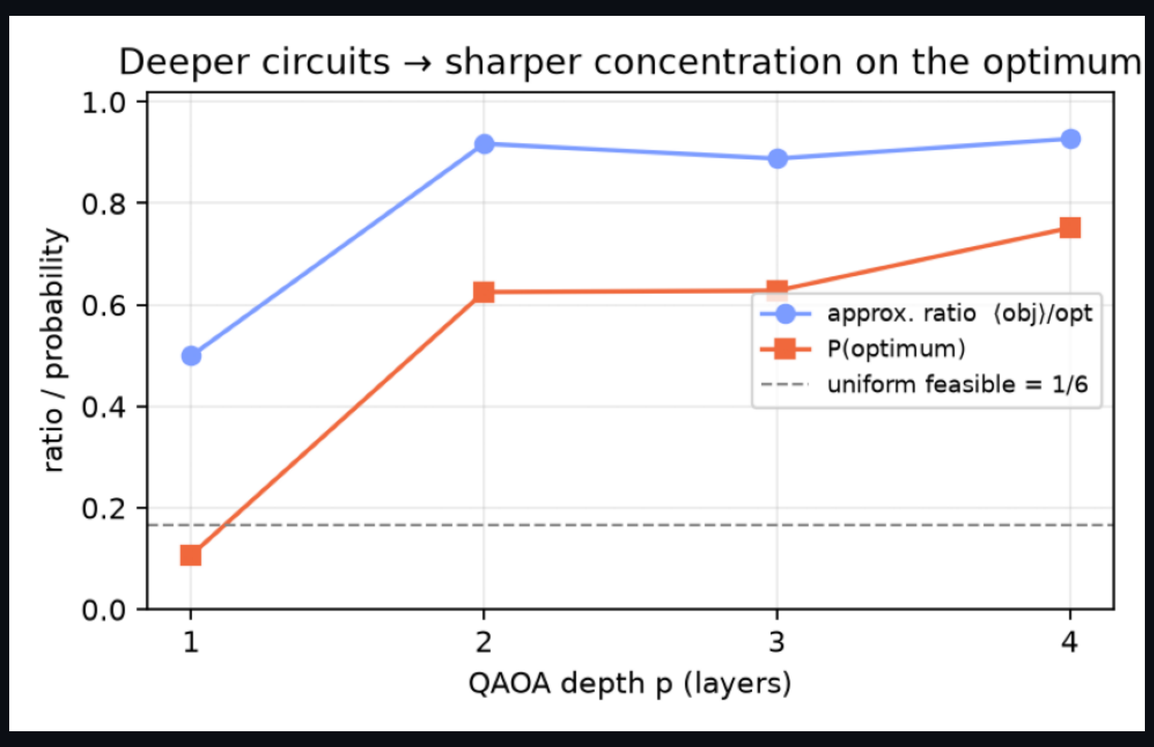
QNLPAI 2026 · Quantum Computing for AI — Quantum Optimization (QNLP bridge) · QLPC Project

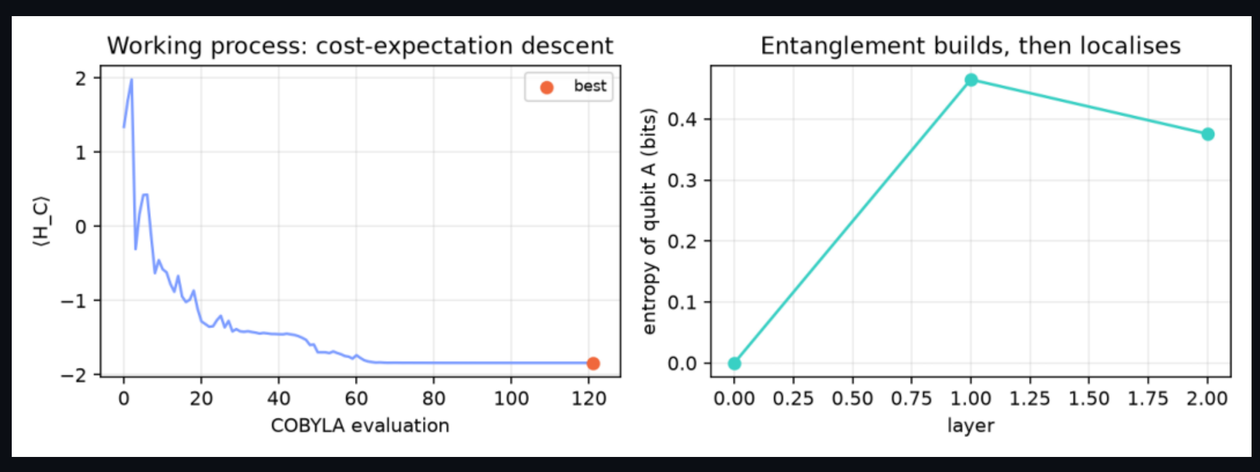
PROBLEM QAOA is reported as a black box — one bitstring, one ratio — hiding HOW the state decides, with no port for the language-shaped views that drive real allocation.

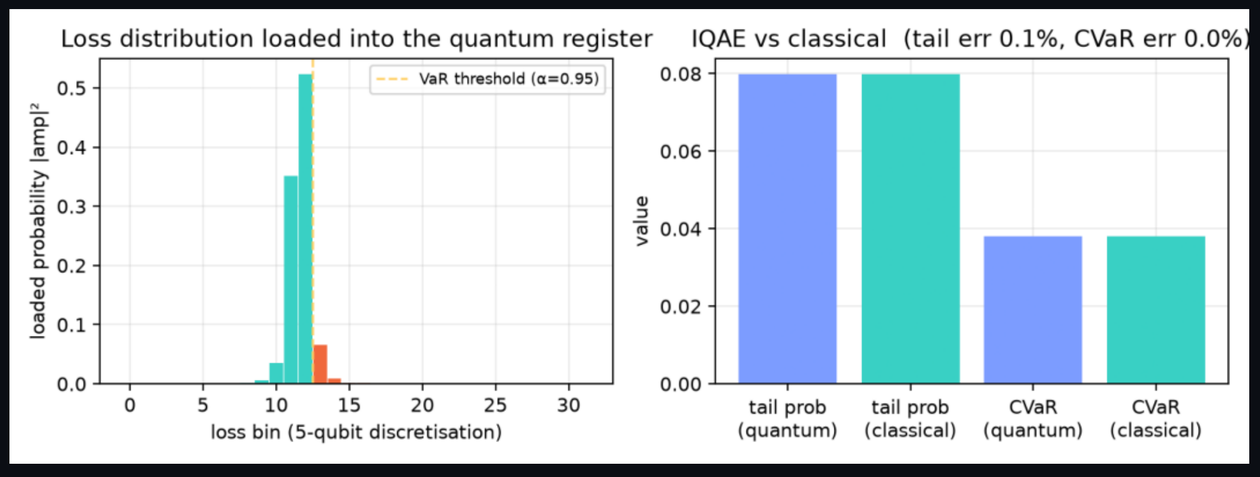
APPROACH Instrument the quantum core of QLPC on IBM Qiskit so every state is observable: (1) snapshot the statevector each QAOA layer; (2) encode an LLM natural-language view as a tilt on expected returns that deforms the cost Hamiltonian; (3) IQAE for tail / CVaR risk.











RESULTS (exact statevector, fixed seeds)

- State concentrates on the optimum: P(opt) 0.062 → 0.63 at p=2 = 10.0× a uniform guess; approx-ratio 0.92; exact optimum recovered by best sample.
- **Language steers the ground state: optimum {A,B} → {A,C} → {C,D}** QAOA modal selection tracks it in all 3 regimes.
- IQAE tail-prob within 0.07% of the loaded distribution (the quantum-estimated quantity).

HONEST LIMITS small, well-separated, simulated instances; no scaling/supremacy claim; additive tilt.

Reproducible: `python -m paper.experiments` (Qiskit 2.x + Aer) · circuits.py · experiments.py · results.json